

Chapter 1: History of Engineering Practice

- **1.1 Engineering and Its History**

1. From the History of engineering profession, what lessons the freshly graduated Engineers would learn. [5] (2079 Jestha, Q1)

- **1.2 Man and Society**

1. Define the term society and community. [1] (2080 Chaitra, Q1)
2. Explain the relationship between human and society. [2] (2079 Chaitra, Q1a)
3. Define society. Illustrate elements of society. Describe the relationship between man and society. [Part of 10] (2072 Magh, Q1)
4. Why is society necessary to a man, or to an engineer? Explain. [4] (2076 Bhadra, Q1)
5. Why are men and society so important to engineering profession? [4] (2073 Magh, Q1a)
6. Define Society. Explain briefly about the types of society in the sociological point of view. [4] (2079 Shrawan, Q1a)
7. Explain criteria of society. [4] (2074 Magh, Q1)
8. Define society. What are the fundamental social needs to be addressed for the survival of every type of society? [4] (2071 Magh, Q1a)
9. What are the major activities to be governed by the society for its survival? [4] (2073 Bhadra, Q1a)
10. Write down in brief the Characteristics Features of Society. What are the elements of Community. [5] (2074 Bhadra, Q1a)
11. What are the differences between community and society? Briefly describe the theories of social change. [2+3] (2069 Bhadra, Q1a)
12. Define Society and Social Change. [2] (2077 Chaitra, Q1)
13. What is social change? What are the factors causing social change? [Part of 8] (2072 Ashwin, Q1)
14. List down and explain the various theories of social change with example. [1+3] (2081 Shrawan, Q1)
15. Explain the conflict theory of social change and its relevance in Nepal. [1+3] (2078 Kartik, Q1a)
16. Explain what type of Social change theory is suitable for your society. [4] (2075 Bhadra, Q1)
17. Write short notes on: Society and development. [2] (2073 Bhadra, Q5e)

- **1.3 Individual Freedoms vs. Societal Goals**

1. Explain how individual freedom balances societal goals. [4] (2073 Bhadra, Q1b)
2. Why the conflict exists between an individual's freedom and the social goal? [5] (2070 Magh, Q1a)

- **1.4 Key Roles of Engineers in the Development Activities**

1. Explain in brief the role of engineers in Social Change. [3] (2080 Chaitra, Q1)
2. Explain why society is important for engineers? What are the key roles that an engineers plays in the society? [4] (2078 Chaitra, Q1a)
3. Why is society necessary for engineers? What are the roles that an engineer can play in the society? [5] (2070 Bhadra, Q1a)

- **1.5 Technology and Social Change**

1. Describe about the impact of technology into the society. [3] (2079 Chaitra)
2. Discuss on the impact of technology into the society. [4] (2071 Bhadra, Q1a)
3. What are the impacts of technology and its changes on society in this 21st century? [5] (2069 Bhadra, Q1b)
4. How does digital technology help our society in economic development? [2] (2077 Chaitra, Q1)
5. Describe the role of technology in social change. [Part of 8] (2072 Ashwin, Q1)
6. Write short notes on: Technology and society. [4] (2079 Shrawan, Q5b)
7. Write short notes on: Technology and society. [2] (2073 Magh, Q5a)

- **1.6 Impact of Computers on Society**

1. Illustrate the impacts of computer on Nepalese society. [Part of 4] (2073 Bhadra, Q1a)
2. Write short notes on: Society and Information Communication Technology (ICT). [5] (2079 Chaitra, Q5a)
3. Write short notes on: Impact of computers in society. [2] (2072 Ashwin, Q5a)
4. Write short notes on: Computer crimes [2.5] (2072 Magh, Q5a)

- **1.8 Engineering Practices in Nepal**

1. In your opinion, what types of technology/ies Nepal should focus on, to develop the rural infrastructure rapidly. Be specific to your area of study/program of your study. [4] (2078 Kartik, Q1b)
 2. Discuss about the engineering practice in Nepal. [4] (2071 Bhadra, Q1b)
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Chapter 2: Profession & Ethics

- **2.1 Profession**

1. Define profession. [1] (2080 Chaitra, Q2a)
2. Define profession. [1] (2077 Chaitra, Q2a)
3. How do you define a Profession? [Part of 8] (2075 Bhadra, Q2)
4. Define Profession. Explain the features of profession briefly. [4] (2078 Kartik, Q2a)
5. Define profession. Illustrate elements of profession. [4] (2076 Bhadra, Q2)
6. What do you mean by Profession? Explain its characteristics. [4] (2073 Bhadra, Q2a)
7. Define Profession. Illustrate the characteristics of a Profession. [5] (2071 Magh, Q2a)
8. Write down the fundamental characteristics of profession. [Part of 4] (2073 Magh, Q2a)
9. What is profession? [Part of 8] (2072 Ashwin, Q2)
10. What is a profession? [Part of 5] (2070 Magh, Q2a)

- **2.2 Engineering Profession**

1. How do you describe engineering profession? Explain the significant features of profession. [2+3] (2069 Bhadra, Q2a)
2. What are the principles of engineering profession. [Part of 8] (2075 Bhadra, Q2)
3. Explain the perception of general public towards Engineer and their professional works in the context of Nepal. [2] (2081 Shrawan, Q2a)
4. Write short notes on: Impact of engineering profession. [2.5] (2069 Bhadra, Q5a)

- **2.3 Professional Association**

1. Define professional association. Write down the objectives of professional association. [1+2] (2080 Chaitra, Q2a)
2. Define professional association. Discuss the role of Nepal Engineers' Association (NEA) for engineering professional development in Nepal. [1+2] (2077 Chaitra, Q2a)

3. How do NEA and NEC contribute to maintain the profession to remain ethical? [Part of 5] (2079 Chaitra, Q1b)
4. Write down the professional code of conduct issued by Nepal Engineering Council (NEC). Elaborate the professional benefit you will get from NEA. [4] (2076 Bhadra, Q8)
5. Recently an engineer was assaulted publicly by some group of people in Terai, Nepal. Explain the role that Professional Associations should play so that such incidences does not repeat again. [4] (2078 Kartik, Q2b)
6. Recently Nepal Engineer's Association (NEA) celebrated its golden jubilee years 2069. In this regard, are you satisfied with role played by NEA in enhancing and upgrading engineering profession and welfare of the engineering till date or not? Justify your thoughts. [5] (2069 Bhadra, Q3a)
7. Write short notes on: Nepal Engineers Association. [2] (2073 Bhadra, Q5d)
8. Write short notes on: Professional institutions. [2] (2072 Ashwin, Q5c)

• 2.4 Ethics

1. What do you understand by the term Ethics? [Part of 4] (2078 Chaitra, Q2a)
2. Define ethics. [2] (2077 Chaitra, Q2b)
3. What is ethics? [Part of 4] (2073 Bhadra, Q2b)
4. What do you understand by ethics? [Part of 5] (2070 Bhadra, Q2a)
5. What is meant by moral and ethics? [Part of 5] (2079 Chaitra, Q1b)
6. Define moral and immoral actions. [2] (2081 Shrawan, Q2a)
7. Why is ethics and moral necessary in engineering profession? [4] (2079 Jestha, Q2)
8. Define Ethics. What is moral and non-moral action? [Part of 5] (2074 Bhadra, Q2b)

• 2.5 Code of Ethics

1. Write in short the code of ethics for engineering profession. [Part of 4] (2073 Bhadra, Q2b)
2. Describe the code of ethics for engineers. [Part of 8] (2072 Ashwin, Q2)
3. Write down the code of conduct to be followed by Engineerings? [Part of 5] (2070 Magh , Q2a)
4. List out the core ethical values to develop code of conduct for professionals. [Part of 4] (2073 Magh)
5. What are the factors which can affects your moral while practicing your profession. What are the core ethical values to prepare code of ethics. [4] (2076 Bhadra, Q6)

6. Why is it necessary for a professional to follow code of conducts? [Part of 4] (2078 Chaitra, Q2a)
7. Why are code of conducts required for professionals? [Part of 5] (2070 Bhadra, Q2a)
8. What are the codes of ethics for engineers according to Nepal Engineering Council? [4] (2071 Bhadra, Q2b)

- **2.6 Engineering Professional Ethics**

1. Explain engineering morals, ethics and professionalism. [8] (2074 Magh, Q2)
2. Define Professional ethics. Why do Engineers behave unethically? Explain the role of Nepal Engineering council to control the immoral act of engineers in the Nepal. [1+1+2] (2078 Kartik, Q3a)
3. Write briefly the perspectives of Professional ethics. [Part of 5] (2074 Bhadra, Q2b)
4. How do you differentiate professional ethics with moral behavior? [Part of 4] (2071 Magh, Q2b)
5. What are Factors affecting the morale of a professional engineer? Describe in brief to justify and satisfy yourself in the context of Nepal, to have and maintain the professional and moral ethics. [2+3] (2069 Bhadra, Q2b)

- **2.7 Moral Dilemma and Ethical Decision Making**

1. Explain moral dilemma with examples. Write down and explain priority while making ethical decision briefly. [0.5+3] (2080 Chaitra, Q2b)
2. How does a moral dilemma occur? What are the bases to solve moral dilemma? [4] (2071 Bhadra, Q2a)
3. Define moral dilemma. What are fundamental laws of ethics to overcome moral dilemma? [4] (2071 Magh, Q1b)
4. How do you mean by moral dilemma? Explain various ways to overcome dilemmas. [5] (2070 Magh, Q2b)
5. What are the secular systems or methods of making ethical proper and good decisions? [1] (2081 Shrawan, Q2b)
6. Write and explain the bases of making ethical decision. [6] (2079 Jestha, Q2)
7. Write short notes on: Conflict of interest in profession. [5] (2079 Chaitra, Q5c)

- **2.8 Relation of Engineer with Client, Contractor & fellow Engineers**

1. What do you understand by professional practice? Describe the relation of engineer with client, contractor and fellow engineers. [10] (2072 Magh, Q2)

- **2.9 Duties of an Engineer and Architect**

1. What are the duties and Liability of an Engineer and Architect? [5] (2074 Bhadra, Q1b)
2. What are the detailed duties of an engineer in the profession? [5] (2070 Bhadra, Q2b)
3. Write short notes on: Detailed duties and liabilities of an engineer and architect. [2] (2073 Bhadra, Q5c)

- **2.11 Negligence, Tort and Liability**

1. Define negligence, tort and liability. [3] (2081 Shrawan, Q2b)
2. Explain negligence, tort and liability with examples. [2] (2077 Chaitra, Q2b)
3. What are liability and negligence? [Part of 5] (2070 Magh, Q2a)
4. Explain tort and liability. [Part of 8] (2072 Ashwin, Q2)
5. What do you understand by Negligence, Tort and Liability. What are the elements of Negligence. [5] (2074 Bhadra, Q2a)
6. Explain tort. What are the reasons behind introducing tort law in the society? [1+1] (2069 Bhadra, Q4a)
7. Define vicarious liability. [Part of 4] (2073 Magh, Q2a)
8. Explain vicarious liability with examples. [0.5] (2080 Chaitra, Q2b)
9. Differentiate tort liabilities with vicarious liabilities. What are the conditions necessary to prove professional negligence in engineering practice? [4] (2079 Shrawan, Q2a)
10. Write short notes on: Liability and Negligence. [4] (2078 Chaitra, Q5b)
11. What kind of liability that engineers most suffer from? Explain. [5] (2070 Bhadra, Q3b)
12. Write short notes on: Liability and Negligence. [4] (2071 Magh, Q5c)

Chapter 3: Professional Practice in Nepal

- **3.1 Public Sector Practice**

1. What are the sectors that engineers in Nepal can practice their profession? [5] (2079 Jestha, Q1)

2. What do you understand by practical engineering profession? List down at least 5 organizations each in public and private sector. [4] (2073 Magh, Q3a)
3. Write down and explain the job description of a freshly graduated engineer working in public sector in the context of Nepal. [4] (2081 Shrawan, Q3)
4. Write down and explain the job description of a freshly graduated engineer working in private sector in the context of Nepal. [4] (2080 Chaitra, Q3)
5. What are the job-description of a freshly graduated engineer in a public sector of Nepal? [5] (2079 Jestha, Q3)
6. Write down and explain the job description of a fresh engineering graduate appointed to a public organization. [4] (2078 Chaitra, Q1b)
7. What are the detail job responsibilities of a gazette third class officer regarding engineering profession? [4] (2077 Chaitra, Q3)
8. What may be the job description of a freshly graduated engineer in a public sector? Explain. [5] (2075 Bhadra, Q3)
9. Explain general job description of engineers working in the public sectors. [4] (2074 Magh, Q3)
10. Write Job Description for a freshly graduated engineer in government sector. [4] (2073 Magh, Q1b)
11. What are the job description of a fresh engineer that can be appointed in a public organization. [5] (2070 Bhadra, Q1b)

- **3.2 Private Sector Practice**

1. Write down the job description of an engineer appointed to a private sector. [4] (2079 Shrawan, Q1b)
2. What is a job description of a freshly graduated engineer in a private sector? [4] (2076 Bhadra, Q9)

- **3.3 Job Descriptions of Fresh Engineers**

1. Write down job description of an engineer working in private and public sectors. [5] (2079 Chaitra, Q2b)
2. Write short notes on: Job description of fresh graduates. [2] (2073 Bhadra, Q5b)
3. Write short notes on: Job description of fresh engineers in public and private sector [2.5] (2072 Magh, Q5c)

4. Differentiate public and private sectors for engineering practice. [Part of 10] (2072 Magh, Q1)

- **3.4 Community Participation in Development Activities**

1. Why participatory approach is necessary in development activities? What are their benefits? Explain in the context of Nepal. [4] (2078 Kartik, Q3b)
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Chapter 4: Contract Management

- **4.1 Methods of Work Execution Procurement**

1. List down various method of work execution. [2] (2080 Chaitra, Q4a)
2. Describe in brief the methods of contracting. [4] (2076 Bhadra, Q4)
3. Explain briefly procurement types (Sealed bidding, sealed quotation and direct purchase) of work as per public procurement Act 2063. [4] (2079 Shrawan, Q3b)

- **4.2 Contract**

1. What is contract? Describe briefly the elements of contract. [2+2] (2077 Chaitra, Q4a)
2. What is contract document? What are the essential elements of contract? Explain. [1+3+4] (2075 Bhadra, Q4)
3. What is contract? Describe essential elements of contract. [Part of 8] (2072 Ashwin, Q3)
4. What is contract? What are the essential elements of contract? [5] (2070 Magh, Q3a)
5. Define Contract. What are the essentials elements in the Valid Contract? [5] (2074 Bhadra, Q3b)
6. Describe briefly the elements of contract. [4] (2076 Bhadra, Q3)
7. Describe briefly the elements of contract? [4] (2071 Bhadra, Q4a)
8. Define contract. What are the essential elements of contract? In which condition contract will be void? [4] (2078 Chaitra, Q3a)
9. Explain any four elements of Contract. Differentiate between void and voidable contract. [2+2] (2081 Shrawan, Q4a)
10. Elaborate the differences between void and voidable contract. Explain briefly about general conditions of contract and its essential contents. [2+3] (2069 Bhadra, Q3b)
11. Differentiate between contract and agreement. Describe the importance of contract and explain the elements of contract. [2+3] (2079 Chaitra, Q3a)

12. Explain any three mostly practiced types of contract briefly. [2] (2080 Chaitra, Q4a)
13. Define contract document. What are the priorities of contract documents? Explain the rules of contract interpretation. [4] (2071 Magh, Q4a)
14. How the contract can be interpreted in case of ambiguity? Define percentage contract and in which category of procurement, this form of contract is adopted? [Part of 4] (2073 Bhadra, Q3a)
15. As an Engineer, how does knowledge of contract help you in your professional career? Explain. [2] (2081 Shrawan, Q4b)
16. Write short notes on: General condition of contract. [2] (2073 Magh, Q5c)

• **4.3 Tendering (Bidding) Process**

1. Define tender and tender notice. Also explain their process. [2+2] (2077 Chaitra, Q4b)
2. Define tender. Why is tendering so important? List the essential information to be furnished while a tender notice is published. [4] (2079 Shrawan, Q3a)
3. Describe about the tender and explain the purpose tender. List the essential information to be enclosed in the tender notice while publishing for public purpose. [4] (2078 Chaitra, Q3b)
4. Define tender. Explain the purpose of tender. List the essential informations to be contained in the tender notice. [5] (2070 Bhadra, Q3a)
5. Define Tender, what are the essential informations to be given in the Tender Notice? [4] (2071 Bhadra, Q4b)
6. Write the information required to provide in a tender notice. [5] (2079 Jestha, Q3)
7. What are the information you need to provide in a Tender-Notice? [4] (2076 Bhadra, Q10)
8. You are working in a government office and your boss asked you to prepare a notice for a building construction of Rs 10 corer, how will you prepare a notice and explain all the contents that should be included in the notice to publish? [5] (2079 Chaitra, Q3b)
9. What are the preparations to be made before inviting competitive bidding notice? [Part of 4] (2073 Bhadra, Q3a)
10. Differentiate between bid validity period and bid security validity period. [2+2] (2081 Shrawan, Q4b)
11. What do you understand by Tendering? Why is the wall / services required tendering in government sector? [4] (2073 Magh, Q3b)
12. You are project manager of a consulting firm in XYZ project in Nepal situated in Jumla District. Prepare a Tender notice inviting interested and eligible contractor to construct a

temporary office building for your office with approximate cost of Rs 2,00,00,000. [4] (2080 Chaitra, Q4b)

13. Explain tendering process and contract agreement. [Part of 8] (2072 Ashwin, Q3)

14. Write short notes on: Tender and elements of tender [2.5] (2072 Magh, Q5b)

- **4.6 Prequalification**

1. Write short notes on: Benefits of prequalification [2.5] (2069 Bhadra, Q5c)

- **4.7 Recruitment of Consultants**

1. What are the methods of recruitment of consultant? Explain the purpose of Expression of Interest (EOI) and Request for Proposal (RFP). [2+2] (2078 Kartik, Q4a)

2. What are the methods of recruitment of consultant. Explain the purpose of EOI and RFP. [5] (2074 Bhadra, Q3a)

Chapter 5: Regulatory Environment

- **5.1 Nepal Engineering Council**

1. Differentiate between NEA and NEC. [2] (2080 Chaitra, Q5)

2. “You should not practice Engineering without acquiring a license.” Justify this statement. [4] (2079 Shrawan., 4a)

3. What are difference between NEA and NEC? [2] (2075 Bhadra, Q5)

4. Explain engineering council code of conduct and council act 2055 BS. [6] (2074 Magh, Q4)

5. Write short notes on: Nepal Engineering Council [2.5] (2069 Bhadra, Q5b)

6. Write short notes on: Jurisdiction of Nepal Engineering Council. [2] (2073 Bhadra, Q5a)

7. Write short notes on: Scope of Nepal engineering council (NEC). [2] (2073 Magh, Q5b)

8. Write shot notes on: scope of Nepal engineering council [2.5] (2072 Magh, Q5d)

9. What are the detail duties and responsibilities of Nepal Engineering Council (NEC)? [Part of 4] (2071 Magh, Q2b)

- **5.3 Intellectual Property Right**

1. What is intellectual property right? [Part of 4] (2073 Bhadra, Q3b)

2. What is an intellectual property right? [Part of 4] (2073 Magh, Q4a)

3. What are intellectual property right and why is it necessary? [5] (2079 Jestha, Q4)

4. Define Intellectual Property Right. [1] (2078 Kartik, Q4b)
5. Describe intellectual property rights, copy rights, trademark and patent rights. [Part of 10] (2072 Magh, Q3)
6. Define intellectual property. Explain briefly trademark, and copyright in detail. [1+4] (2079 Chaitra, Q4b)
7. Define intellectual property. Explain trademark, service mark, patent and copyright in detail. [8] (2077 Chaitra, Q5)
8. Define intellectual property. Explain trademark, service mark, patent and copyright in brief. [4] (2076 Bhadra, Q5)
9. Differentiate Patent right and Trademark right. [Part of 4] (2071 Bhadra, Q5a)
10. Explain copy right act, patient right and trademark. [6] (2074 Magh, Q5)
11. How do you distinguish copyright with patent right? [3] (2069 Bhadra, Q4a)
12. Define intellectual property rights. Explain (i) the process of registering copyrights in Nepal and (ii) the penalty for violating copy right in Nepal. [1+2+2] (2081 Shrawan, Q5)
13. Define copy right, patent right, design right and Trademarks. Explain briefly the punishment of violating copy right act in Nepal. [2+2] (2078 Kartik, Q4b, OR option)
14. Explain the Intellectual proportion and also explain the procedure of registration of copy right. [5] (2070 Magh, Q4b) *Note: "Intellectual proportion" likely a typo for "Intellectual Property."*
15. Write short notes on: Intellectual property right. [4] (2078 Chaitra, Q5c)
16. Write short notes on: Trade mark [2.5] (2069 Bhadra, Q5d)
17. Write short notes on: Trade marks, Patents, Design. [2] (2073 Magh, Q5e)
18. Write short notes on: Copyrights, Patent and Trademarks. [4] (2071 Magh, Q5a)

• 5.5 Building Codes and Bylaws

1. How can building codes help the people to be safe? Explain. [5] (2070 Magh, Q1b)
2. Write short notes on: Building codes and by laws. [2] (2072 Ashwin, Q5e)

• 5.6 Introduction to Labour Act, Nepal

1. Write in briefly the working hour provision of labour law. [Part of 4] (2073 Bhadra, Q3b)
2. Explain about the labour law and explain why we should be aware of labour law while working in public or private organisation's. [3+2] (2079 Chaitra, Q4a)
3. What are the general principle of leave and holidays? Explain the steps of disciplinary action for misconduct, as per the Labor Act-2074 of Nepal. [4] (2078 Chaitra, Q4a)

4. Explain various provisions regarding working hours as well as health and safety provisions according to labor law. [2] (2080 Chaitra, Q5)
5. Describe the health and safety provision of labour law? [5] (2070 Magh, Q3b)
6. Why should you be aware of Labour Law? What will you do if a construction worker dies by falling from height in your site? [5] (2071 Magh, Q4b)
7. Write short notes on: Labor Law [4] (2079 Shrawan, Q5a)
8. Explain contract law and labour law in Nepal. [8] (2074 Magh, Q7)
9. Explain business law and labour law in Nepal. [4] (2073 Bhadra, Q4a)
10. Write short notes on: Labour and business law. [2] (2072 Ashwin, Q5b)

- **5.7 Business Organizations**

1. Write down the characteristics of company business organization. What are the sources of business law in Nepal? [4] (2071 Bhadra, Q5a)
2. Differentiate between public and private company. [Part of 4] (2073 Bhadra, Q3b)
3. Differentiate between public limited company and private limited company. [5] (2069 Bhadra, Q4b)
4. Write short notes on: Types of business enterprises. [5] (2079 Chaitra, Q5b)

- **5.8 Company Registration in Nepal**

1. Explain the process of registering and company in Nepal briefly. [3] (2078 Kartik, Q4b)
2. What are the necessities of a company registration in Nepal? [3] (2075 Bhadra, Q5)
3. Write company registration procedure in Nepal. [Part of 4] (2073 Magh, Q3a)
4. Write short notes on: Company registration process [2.5] (2072 Magh, Q5e)
5. Write short notes on: Registration Process of engineering consulting firm. [2] (2073 Magh, Q5d)

Chapter 6: Contemporary Issues in Engineering

- **6.1 Globalization and Cross Culture Issues**

1. Please describe about globalization with suitable examples. [Part of 4] (2079 Shrawan, Q4b)
2. Explain globalization and cross cultural issues. [4] (2074 Magh, Q6)
3. Explain globalization and cross cultural issues. [4] (2073 Bhadra, Q4b)

4. What are the challenges of globalization do you see on engineering profession? Describe. [3] (2081 Shrawan, Q6a)
5. Write down the merits and demerits of globalization for developing countries like Nepal. [Part of 4] (2078 Chaitra, Q4b)
6. Define globalization. [Part of 4] (2073 Magh, Q4b)
7. Globalization has been an eye open opportunity for developing country. It helps transfer technology and development process but also a culture that loosens confidence on nationality. How can people work for development within their own culture with others as guide? Explain. [10] (2070 Bhadra, Q4)

- **6.2 Public-Private Partnership (PPP)**

1. Define public private partnership. [Part of 4] (2073 Magh, 4b)
2. What do you understand by the public private partnership in Nepal? [2] (2075 Bhadra, Q6)
3. Define PPP. Explain the importance of private sector in Development. [1+2] (2080 Chaitra, Q6a)
4. Describe about public private partnership (PPP) with suitable examples. [Part of 4] (2079 Shrawan, Q4b)
5. How the PPP models help in any development activities? [Part of 4] (2078 Chaitra, Q4b)
6. What are the risk factor in public private partnership projects? [Part of 4] (2076 Bhadra, Q7)
7. How the PPP model help in any development activities? What is its significance in a developing country like Nepal? [4] (2071 Bhadra, Q5b)
8. Explain in detail the "public private partnership". [Part of 10] (2072 Magh, Q3)
9. Write short notes on: Public Private Partnership. [4] (2071 Magh, Q5b)

- **6.3 Safety, Risk and Benefit Analysis**

1. Describe safety risk and benefit analysis in engineering practice. [Part of 10] (2072 Magh, Q4)
2. Elucidate the objective of safety engineering. [Part of 4] (2073 Magh, Q4b)
3. "Engineering profession always emphasizes on safety first in engineering delivery" Describe with typical examples. [8] (2072 Ashwin, Q4)
4. Write short notes on: Safety, Risk and Benefit analysis [4] (2079 Shrawan, Q5c)

- **6.4 Conflict and Dispute Management**

1. What is conflict? What are the causes of conflict and methods to mitigate them? [4] (2077 Chaitra, Q6)
2. Define dispute and conflict. Explain briefly various methods of resolving conflict. [1+2] (2080 Chaitra, Q6b)
3. What are the methods to resolve conflict? [3] (2075 Bhadra, Q6)
4. What are the methods to resolve dispute? [Part of 4] (2076 Bhadra, Q7)
5. How do you resolve conflicts and disputes arising in professional practice? [Part of 10] (2072 Magh, Q4)
6. What is conflict? Discuss the dispute resolution process? [5] (2070 Magh, Q4a)
7. Why is Dispute Management in any project necessary? Write its reasons and your solutions. [5] (2079 Jestha, Q5)
8. What are levels and sources of conflict in the organization? [Part of 4] (2073 Magh, Q4b)
9. Write short notes on: Conflict and Dispute management. [4] (2078 Chaitra, Q5a)
10. Write short notes on: Conflict and dispute management. [2] (2072 Ashwin, Q5d)

- **6.5 Development and Environment**

1. “Construction is Destruction to Environment”. Do you agree with this statement? Give reasons. (2081 Shrawan, Q6b)

Chapter 7: Case Studies

- **7.1 Code of Ethics**

1. Shashikala is working as an engineer in municipality. One of the road construction project to be implemented by her office is surveyed and designed by a consulting firm that is owned by her college friend Mr. Pran. Mr. Pran had contracted with the municipality office to survey and design the road through open competitive bidding system. While executing the survey work Shashikala convinced Pran to change the road alignment through her land that would increase the value of her land but would not benefit public so far. The project designed by Pran was presented at municipal office in presence of concerned employees including Shashikala and project was approved. Complain against it has been filed for misuse of Shashikala's authority and making loss of public property. As a member of investigation committee on this complain, whom would you declare as guilt and why – Engineer Shashikala or owner of consulting firm, Pran? [5] (2081 Chaitra, Q7)
2. You are working as an engineer from a consulting firm in Gorkha. Your friend Mr. John is working as a project engineer from contractor in the same project. Mr. John invites you for dinner every Saturday to celebrate a weekend and request you to share the guest house for your accommodation. What would you do? Prepare a case study based on code of ethics. Make supplementary assumptions if necessary. [4] (2078 Chaitra, Q2b)
3. You are working as an Engineer from a Consulting Engineering Firm. Your contractor friend Er. Hari is working as a Project Engineer from contractor office in the same project. Mr. Hari invites you for dinner every Friday to celebrate "good Friday" and request you to share the guest house for your accommodation. What would you do in maintaining the code of ethics and friendship relation? [8] (2071 Magh, Q3)

- **7.2 Moral Dilemma**

1. A factory was using a chemical in processing a product. The storage tank of chemical waste (which was hazardous to health and the environment) had a leak. During inspection, an engineer 'A' came to know that the leakage already taken place might cause adverse impact on health and hygiene of the surroundings. Mr. 'A' informed this to his boss about the event. Considering the possible social objection, the boss requested Mr. 'A' to be silent on the issue and also hinted to Mr. 'A' that he may have to be out of the job if the case would go into the

hands of social reformers. If you are Mr. 'A', what you would do? Discuss. [5] (2079 Chaitra, Q2a)

2. Recently the MPCD arrested few engineers of different stream on charge of producing and possessing fake engineering degrees and certifications. It is also not the case that the arrested engineers did not go to any engineering college. They did go to recognized engineering colleges and had attended the classes regularly during their academic course but had failed in few semester exams. Despite their repeated attempts, they were unable to overcome the fail results resulting harsh situation facing barrier for their career further education and professional development. In the meantime, they decided to get fake degrees and certifications to eliminate severe situations. Before arrest, they were working in esteemed company with recognizable performance. How do you judge the behavior/act of arrested engineers on which background? What will be your suggestion to mitigate such situation/incidents never happen again in the society. [4] (2073 Magh, Q2b)

- **7.3 Negligence and Liability**

1. Mr. Ronaldo, an engineering graduate, carried out survey for telephone line requirement for a newly growth city near Lalitpur district. He estimated 1000 lines for five year period. But in 3 years period, the demand for telephone line year dramatically increased and application for telephone line was reached 2000. Discuss this case reference to reasonable skill and competency of a professional. [4] (2079 Shrawan, Q2b)
2. Engineer as a political leader pressurize a contractor for low quality construction and bargains with contractor for unwanted payment. Write down for and against it. [4] (2077 Chaitra, Q7)
3. ABC computer consulting Pvt. Ltd was awarded a contract to install a WAN/LAN connection in XYZ Pvt. Ltd. Ram a computer engineer from ABC, went to install the WAN/LAN in XYZ. In the meantime, he was unsatisfied with his job, so installed WAN/LAN at XYZ negligently. After few weeks, manager of XYZ informed to manager of ABC that the installed system is not functioning properly. In response, Hari another engineer from ABC went back to check the installed system and found that Ram's mistake was there. It will cost NRs. 2 lakhs to maintain the system again. ABC claimed but XYZ refused to pay. Discuss who is to blame on which background. [5] (2075 Bhadra, Q7)
4. An Engineer designed a RCC Bridge on behalf of consultant. A class contractor constructed the bridge. After the completion of the construction, traffic was allowed. Six month of after operation, cracks developed in the bridge. A probe team was established by road department

and you are a member of that team. During investigation it was found that the design procedure was okay, but the quality of steel material used was not duly tested. The contractor argued that the procedure of construction was in accordance with the instruction of engineer and specification, however, the workmanship was found not as per specification. There was also lack of proper supervision by the consultant. The design load for the bridge was 20 tons. It was also reported that there happen to pass more than 20 tons vehicles also. The consultant was good friend of contractor. Being a member of probe team, what is your judgment on the failure of this bridge? Explain. [8] (2078 Kartik, Q5)

- **7.4 Contract Disputes**

1. A RCC Bridge was designed by the Designer on behalf of Consultant. This was Constructed by the reputed "A" Class contractor. After the Completion of the Construction, traffic was allowd on the bridge. After Six Months of operation there were crack in the Bridge. A Probe Team was established by Road department. The Design Procedure was Okay, but it was found that quality of steel material used was not duly tested. The Contractor argued that the procedure of construction was in accordance with the instruction of Engineer and specification. There was also lack of proper supervision by the Consultant. The Design load for the Bridge was 20 tons. It was also reported that there happen to pass more than 20 tons vehicles also. The Consultant was good friend of Contractor. Being a member of Probe Team, what is your judgment on the failure of this bridge? [10] (2074 Bhadra, Q4)
2. Sub-contractor Mr. A was severely beaten by his labors because he had not paid them for more than seven months. During investigation it was revealed that client did not pay to the contractor and contractor did not pay to the subcontractor. The client didn't pay because consultant did not submit the necessary bill report. The consultant says the work is defective and quality of the work is not as per specifications. While the contractor does not agree because it was not informed on site and blames that the consultant wants covert money. Discuss the case considering contract law, tort, liability and negligence whatsoever applicable. [8] (2071 Bhadra, Q3)

- **7.5 Intellectual Property Rights**

1. Recently Samsung Korea was asked to pay compensation of billions of dollar to Apple Inc. by an American court. What do you know about this case? Are you satisfied with the verdict? Express your opinions clearly. [10] (2069 Bhadra, Q5 OR)

- **7.6 Development and Environment**

1. You are a part of a civil engineering team tasked with constructing a dam in a remote area. The dam will provide irrigation facilities, essential water supply for nearby communities and generate hydroelectric power, boosting regional development. However, the proposed site is within a pristine ecosystem, home to several endangered species and culturally significant sites for indigenous communities. Building the dam will involve clearing large areas of forest, flooding habitats, and potentially displacing indigenous communities. Being a member of a team, discuss the case and recommend your views and opinion in this situation. [6] (2080 Chaitra, Q7)